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September 23, 1964
SUPERSEDING
Fed. Spec. W-F-441
October 31, 1951

FEDERAL SPECIFICATION

FLATIRONS, ELECTRIC

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers electric flatirons with adjustable automatic temperature control.

1.1.1 *Federal specification coverage.* Federal specifications do not cover all types and sizes of the commodity indicated by the title of the specification, but are intended to cover only those most generally used by the Federal Government.

1.2 Classification.

1.2.1 *Types and classes.* Flatirons shall be of the following types and classes as specified in the invitation for bids (see 6.2):

Type I—Alternating-current.

Class 1—Weight 4.5 pounds or less.

Class 2—Weight more than 4.5 pounds.

Type II—Direct-current.

Class 1—Weight 4.5 pounds or less.

Class 2—Weight more than 4.5 pounds.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

QQ-C-320—Chromium Plating (Electrodeposited).

QQ-N-290—Nickel Plating (Electrodeposited).

PPP-B-636—Box, Fiberboard.

PPP-B-843—Cushioning Material, Cellulosic.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following publications form a part of this specification to the extent specified herein. Unless otherwise indicated, the issues in effect on date of invitation for bids or request for proposal shall apply.

Underwriters' Laboratories, Inc., (UL) Standards:

No. 141—Electric Flatirons and Ironing Machines.

No. 498—Attachment Plugs and Receptacles.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, Ill. 60611; 161 Sixth Avenue, New York, 13, N. Y.; or 1655 Scott Lane, Santa Clara, California 95050.)

National Electrical Manufacturers' Association (NEMA) Standards:

DA1–1958—Household Automatic Electric and Steam Irons.

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 155 East 44th Street, New York 17, N. Y.)

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.2), before production is commenced, a sample of the flatiron shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. The approval of the preproduction sample authorizes the commencement of production, but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

3.2 Standard product. Except for any modifications or additional requirements specified herein, the flatirons and component parts covered by this specification shall be the manufacturer's current standard product. All parts shall be new and unused and all flatirons or components furnished under any specific contract shall be physically and mechanically identical.

3.3 Materials. The materials specified shall conform to the applicable specifications and requirements specified hereinafter. Materials not definitely specified shall be of the quality normally used for the purpose in commercial practice.

3.4 Standards.

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3.4.1 *UL Standards.* Unless otherwise specified herein, the flatirons shall conform to UL Standard 141 and the attachment plugs and receptacles to UL Standard 498.

3.4.1.1 *UL certification.* Prior to approval of the preproduction sample, if one is submitted, or prior to approval of the first shipment, the supplier shall submit satisfactory evidence to the contracting officer or his authorized representative that the flatirons he proposes to furnish under this specification meet the requirements of UL Standards 141 and 498. Acceptable evidence of meeting the requirements of this standard shall be the UL label, an approved UL listing, or a certified test report from a recognized independent testing laboratory acceptable to the Government, indicating that the flatirons offered have been tested and conform to the requirements of UL Standards 141 and 498.

3.4.2 *NEMA standard.* The flatirons shall conform to the performance requirements of NEMA Standard DA1–1958 for temperature regulations of the soleplate, the temperature of the handle and thermostatic control, lifetest, actual power input and mechanical ruggedness.

3.4.2.1 *NEMA certification.* Prior to approval of the preproduction sample, if one is submitted, or prior to approval of the first shipment, the supplier shall submit satisfactory evidence to the contracting officer or his authorized representative that the flatirons he proposes to furnish under this specification meet the requirements of NEMA Standard DA1–1958. Acceptable evidence of meeting NEMA requirements shall be the manufacturers' certified data or a certified test report from a recognized independent testing laboratory, acceptable to the Government, indicating that the flatirons offered have been tested and conform to NEMA Standard DA1–1958.

3.5 Design. The flatirons shall be designed to have a voltage rating of 120 volts as specified in 3.5.1 through 3.5.5 and a thermostatic control to regulate the temperature of the soleplate (see 3.4.2).

3.5.1 *Type I.* The type I flatirons shall be designed for use on 120 volt alternating current only.

3.5.2 *Type II.* The type II flatirons shall be designed for use on both 120 volt alternating current and 120 volt direct current.

3.5.3 *Class 1.* The class 1 flatirons shall be designed to have a rated power input of 1,000 watts with a tolerance of plus 5 percent and minus 10 percent.

3.5.4 *Class 2.* The class 2 flatirons shall be designed to have a rated power input of 700 watts with a tolerance of plus 5 percent and minus 10 percent.

3.5.5 *Voltages.* The voltage at which the flatiron shall be designed to operate at rated power input, shall be 120 volts. Rated voltage, the nominal circuit voltage for which the flatirons shall be designed, shall be 120 volts. The range of the impressed voltages shall be a minimum of 105 volts and a maximum of 125 volts.

3.5.6 *Speed of heating.* Each flatiron shall attain a mean soleplate temperature of not less than 400°F. within 8 minutes after being connected to power with the voltage ratings as applicable in 3.5.5. and tested as specified in 4.4.1.

3.6 Construction.

3.6.1 *Handle.* The handle shall be made of phenolic or a similar high temperature composition and shall be molded in one piece with the supporting bracket or shall be securely fastened to the handle bracket.

3.6.2 *Soleplate.* The soleplate shall consist of either aluminum or other alloys, or stainless steel overlay on aluminum or other alloys. The soleplate area shall be not less than 22 square inches. The soleplate front shall come to a distinct point and shall have a double button nook. The bottom surface of the soleplate shall show no variation greater than 0.030 inches from a flat sur-

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face on which the flatiron rests when tested as specified in 4.4.2.

3.6.3 *Safety stand.* Each flatiron shall have a permanently attached safety stand or rest.

3.6.4 *Thermostat.* Each flatiron shall have a manually adjustable thermostat. The thermostat knob or lever shall be of such shape and so located that temperature adjustment can be made without the hand of the operator coming in contact with a hot surface. The type I flatiron shall have a switch incorporated into the thermostat for the "OFF" position.

3.6.5 *Cords.* Each flatiron shall have a cord, not less than 6 feet in length, either permanently attached to or detachable from the flatiron. One end of the cord shall be equipped with an attachment plug, conforming to the Underwriters' Laboratories, Inc. Standard 498, for connection of the cord to the power supply.

3.6.5.1 *Detachable cord.* When a detachable cord is furnished, the flatiron shall be equipped with on-the-iron contact posts of corrosion-resisting material. One end of the detachable cord shall be equipped with a flatiron plug having sleeves of corrosion-resisting material to fit the contact posts on the iron.

3.6.5.2 *Anti-kink device.* Each cord shall have an anti-kink device not less than 1½ inches in exterior length. The device shall be located at the flatiron plug, when the cord is detachable, and at the flatiron when the cord is attached.

3.7 Finish. All outside metallic surfaces of the flatiron except the soleplate shall have a chromium over nickel plating. The plating thickness on ferrous metal outside surfaces shall conform to 3.7.1 and for copper, brass, and bronze outside surfaces to 3.7.2, and shall be tested as specified in 4.3.2.

3.7.1 *Ferrous metal.* The nickel plating on ferrous metal shall conform to class 1, type IV as specified in QQ–N–290. The chromium plating shall conform to class 1, type I as specified in QQ–C–320.

3.7.2 *Copper, brass, bronze.* The nickel plating shall conform to class 1, type VII as specified in QQ–N–290. The chromium plating shall conform to class 1, type I as specified in QQ–C–320.

3.8 Marking. The flatirons shall be marked in accordance with the marking requirements of Underwriters' Laboratories, Inc., Standard 141, for electric flatirons and ironing machines.

3.8.1 *Marking of thermostat control knob.* The thermostat control knob for type I flatirons shall be marked to indicate settings for not less than four fabrics and shall have an OFF position. The fabric settings shall include "Synthetics" or "Wash and Wear", "Wool", "Cotton", and "Linen". The thermostat control knob for type II flatirons shall be marked to indicate not less than four fabric settings as for type I.

3.9 Workmanship.

3.9.1 *Riveted connections.* Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools, and shall completely fill the holes. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.9.2 *Welding and brazing.* All surfaces of parts to be welded or brazed shall be clean. Welding and brazing shall be accomplished in a manner which will prevent the occurrence of burn holes, cracks, fractures or incomplete fusion. All scale or flux deposits shall be removed from finished welds.

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3.9.3 *Electrical connections.* Electrical connections shall be mechanically secured and welded or brazed; except connections on screw type terminals, which shall be mechanically secured.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Preproduction sample inspection. When a preproduction sample is required, it shall be examined in accordance with the defects in table I, and dimensions specified and tested as specified in 4.4.1 thru 4.4.3.

4.3 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL–STD–105, except where otherwise indicated.

4.3.1 *Component and material inspection.* In accordance with 4.1, the supplier is responsible for insuring that components and materials used are manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or, if none, in accordance with this specification.

4.3.2 *Intermediate test.* The plating thickness shall be tested as specified in QQ–N–290 for nickel plating and QQ–C–320 for chromium plating, to determine compliance with the plating requirements specified in 3.7.1 and 3.7.2, as applicable.

4.3.3 *End item inspection.* A lot shall consist of all flatirons of the same type and class offered for acceptance at one time. The sample unit shall be one completely fabricated flatiron.

4.3.3.1 *Visual examination.* Examination of the end item shall be in accordance with the classification of defects in table I. The inspection level shall be level II and the acceptable quality level (AQL) 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I—*Classification of defects*

Examine	Defect	Classification	
		Major	Minor
Finish	Finish not as specified	X	
	Not finished where required	X	
Construction and workmanship, general (applicable	Part missing or not specified type or material.	X	

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TABLE I—*Classification of defects (Continued)*

Examine	Defect	Classification	
		Major	Minor
to components and assemblies)	Fractured, split, punctured, dented, deteriorated, bowed, malformed, or otherwise impaired.	X	
	Misplaced, loose or not in proper alignment.	X	
	Sharp burr, sliver or splinter.	X	
	Operation omitted or not properly performed.	X	
	Functioning component that requires abnormal force to operate.		X
	Not connected or joined as specified or poorly accomplished.	X	
	Soleplate area not as specified.	X	
	Thermostat not as specified.	X	
	Cords not proper length.	X	
	Attachment plugs not as specified.	X	
Welding and brazing	Missing, incomplete, burn holes, cracked or fractured.	X	
	Slag inclusion, slight undercut, not smooth and uniform, scale or flux deposits not removed.		X
Marking	Missing, incomplete, not legible		X

4.3.3.2 ***Dimensional examination.*** *In*spection shall be made of the end item for compliance with dimensions specified. Any deviation from specified requirements shall constitute a defect. The inspection level shall be S–2 and the AQL shall be 4.0, expressed in terms of defects per hundred units.

4.3.3.3 *Testing of the end item.* End item testing shall be performed on the flatirons as specified in 4.4.1 through 4.4.3. The inspection level shall be S–2 with an AQL of 4.0 expressed in terms of defects per hundred units.

4.3.4 *Standards compliance.* The supplier shall make available, for the Government representative, proof of compliance with the requirements of the Underwriters' Laboratories, Inc. Standards and NEMA Standard as applicable to 3.4.1 and 3.4.2.

4.3.5 *Examination of preparation for delivery.* An examination will be made to determine that packaging, packing, and marking as required by section 5 of this specification are complied with. Defects will be scored as specified in table II. The sample unit shall be one shipping container fully prepared for delivery. The lot shall be the number of containers offered for delivery at one time. The inspection level shall be S–4 and the acceptable quality level shall be 6.5, expressed in terms of defects per hundred units.

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TABLE II—*Examination of preparation for delivery*

Examination	Defect
Markings	Missing; incorrect; illegible; of improper size, location, sequence or method of application.
Materials	Any nonconforming component; component missing, damaged or otherwise defective.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, inadequate stapling, bulging, or distortion of containers, strapping or sealing inadequate, improper or missing; closures improperly affixed.
Weight	Net weight exceeds requirements.

4.4 **Tests.**

4.4.1 *Speed of heating.* The flatiron shall be connected to power as specified in 3.5.5, placed on the stand and the thermostat set at the highest temperature setting. Soleplate temperatures shall be taken continuously, and the length of time at which the mean soleplate temperature reaches 400°F. shall be determined in accordance with 3.5.6.

4.4.2 *Variation of the soleplate.* Place the soleplate of the flatiron on a true plane surface and with a feeler gage determine variation of the soleplate in accordance with 3.6.2.

4.4.3 *Weight of iron.* The iron shall be weighed to determine compliance with the weights specified in 1.2.1.

5. PREPARATION FOR DELIVERY

(Fed. Std. No. 102 shall be referred to for definitions and application of the various levels of packaging and packing for civil agency procurement.)

5.1 **Packaging.** Packaging shall be level A, B, or C, as specified (see 6.1).

5.1.1 *Level A or B.* Each flatiron shall be wrapped in cushioning material conforming to type I or II, class B of PPP–C–843. The flatiron shall then be placed in a fiberboard box conforming to type CF or SF, class Domestic, style FOL of PPP–B–636, except that the minimum bursting strength of the fiberboard shall be 200 pounds per square inch. The flatiron shall be immobilized within the fiberboard box with fiberboard inserts of the same material as the box. Each box shall be securely closed in accordance with the appendix of the box specification.

5.1.2 *Level C.* Flatirons shall be packaged in accordance with the industry's practice.

5.2 **Packing.** Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 *Level A.* Flatirons, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to V3c or V3s, style RSC of PPP–B–636. The weight of contents of each shipping container shall not exceed 65 pounds. Each shipping container shall be closed, waterproofed by means of tape, and reinforced with flat steel strapping, or tape banding in accordance with the appendix of the container specification.

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5.2.2 *Level B.* Flatirons, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to type CF or SF, class Domestic, style RSC of PPP–B–636. The weight of contents of each shipping container shall not exceed 65 pounds. All flaps of the fiberboard shipping container shall be securely sealed with an adhesive commercially used for the specific product being packed by application throughout the entire area of contact between the flaps or by a combination of metal stitching the bottom flaps and sealing the top flaps with adhesive.

5.2.3 *Level C.* Flatirons, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 **Marking.**

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military requirements.* In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL–STD–129.

6. NOTES

6.1 **Intended use.** The flatirons covered by this specification are for general use in the normal hand ironing of fabrics.

6.2 **Ordering data.** Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Type and class of flatirons required (see 1.2).

(*c*) When preproduction sample is waived (see 3.1).

(*b*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 **Transportation description.** The transportation description and minimum weights applicable to this commodity are:

Rail:

Sadirons with or without stands, electric.

Carload minimum weight 30,000 pounds.

Motor:

Sadiron not otherwise indexed, with or without stands, electric.

Truckload minimum weight 30,000 pounds, subject to rule 115, National Motor Freight Classification.

MILITARY CUSTODIANS:

Army—GL

Navy—SA

Air Force—69

Interested activities:

Review

Army—GL EL SM

Navy—SA

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User

Navy—MC

Preparing activity:

Army—GL

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